



Monodispersed zinc oxide nanoparticles as multifunctional additives for polycarbonate thermoplastic with high transparency and excellent comprehensive performance

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Abstract

Optical plastic-based organic/inorganic nanocomposites have great application potentials. However, it is still challenging to disperse inorganic nanoparticles into plastic matrices uniformly, especially for thermoplastics with high processing temperatures. We propose a “liquid–solid masterbatches” blending method to completely solve the agglomeration problem of nanoparticles in the thermoplastic matrix and therefore fabricate a highly transparent nanocomposite of zinc oxide (ZnO)/polycarbonate (PC). Liquid masterbatches were prepared based on the monodispersed ZnO nanoparticles in dichloromethane and then spray-dried to form the solid masterbatches. Finally, ZnO/PC nanocomposites were prepared by melting extrusion. ZnO nanoparticles have a uniform particle size of about 6 nm with excellent thermal stability, which is suitable for melt blending with PC at high temperatures. They remarkably improve the ultraviolet (UV)-shielding property, anti-photoaging capacity, and mechanical performance of PC. More importantly, the transparency of the nanocomposites is the same as pure PC, resulting from the homogeneous dispersion of ZnO nanoparticles in the PC matrix.

Keywords ZnO nanoparticles · Nanocomposites · Homogeneous dispersion · Optical properties · Mechanical testing

1 Introduction

Organic/inorganic optical nanocomposites combine the performance advantages of inorganic nanomaterials and organic polymer materials, which have not only excellent characteristics of optical polymer: high transparency, low density, corrosion resistance, and easy processing [1, 2], but also good dielectric or conductivity [3–6], heat transfer [7, 8], wave absorption [9], high-temperature resistance [10], UV resistance [11], and other intrinsic properties of inorganic nanoparticles. Therefore, they have great application

prospects in the fields of saving-energy, electronic information, military, and defense [12–15]. It is driving a technological revolution in the field of new materials and related supporting industries.

Polycarbonate (PC) is a vital engineering thermoplastic [16, 17], which is widely used in the preparation of optical materials and devices. Incorporating inorganic nanomaterials into PC matrix can improve composite performance and endow it with some new functions [18–20]. For example, PC is usually manufactured into transparent windshields, lenses, etc., while it has a poor UV shielding ability, resulting in the damage of objects and people by UV radiation. Adding inorganic ultraviolet absorbers, such as ZnO, CeO₂, and TiO₂, can significantly improve UV resistance [21–23]. However, inorganic nanoparticles tend to aggregate and have poor dispersion in organic polymer matrices due to their large specific surface area and high surface energy, which will cause a decrease in transparency and destroy the performances of optical polymer materials [24].

Combining surface modification of inorganic nanoparticles with solid masterbatch technologies to fabricate thermoplastic-based nanocomposites has usually been adopted to solve the aggregation problems of inorganic

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nanoparticles in polymer matrices [25–27]. The dispersibility of inorganic nanoparticles has been improved notably, but a partial agglomeration in the matrix still exists. Because the preparation of solid nano-masterbatches needs nanopowders as raw materials, nanoparticles synthesized by the liquid-phase precipitation method must be dried inevitably to obtain the nanopowders, and it will cause the aggregations of nanoparticles more or less in the drying process, which cannot be redispersed by shearing stress during melt extruding. As for optical PC materials, any agglomeration of nanoparticles with a size of more than 40 nm will bring about the Rayleigh scattering of light and reduce the transparency of composites [28]. In addition, PC plastic is usually processed by extrusion at a temperature over 250 °C, leading to the decomposition of the modifiers on the surface of inorganic nanoparticles. As a result, the appearance of PC is poor, and properties are decreased [29, 30]. Therefore, it is necessary to develop a novel method to completely solve the agglomeration problem of inorganic nanoparticles in optical thermoplastic matrices. Especially for the optical thermoplastics with high processing temperature, such as PC and polysulfone, the preparation method of nanocomposites can effectively disperse nanoparticles and improve the performances, which also can be used to fabricate the other thermoplastic-based nanocomposites with the excellent properties.

In this work, the reactive precipitation method combined with in situ modification technology and the novel “liquid–solid masterbatch” blending method were adopted to prepare monodispersed ZnO nanoparticles and highly transparent ZnO/PC nanocomposites, respectively. Suitable compound surfactants were chosen to improve the dispersion of ZnO and its compatibility with PC resin. Meanwhile, due to the excellent thermal stability of the selected modifiers, ZnO nanoparticles can fully adapt to the high processing temperature of PC. The aggregation problem of ZnO nanoparticles in PC matrix was solved entirely by the “liquid–solid masterbatches” blending method, contributing to achieving high visible light transmittance and ultraviolet blocking rate of the ZnO/PC nanocomposites. At the same time, the mechanical performances and UV-aging resistance were comprehensively enhanced, and the appearance is unchanged through the high-temperature melt-extrusion process.

2 Experimental details

2.1 Chemicals

Zinc acetate dihydrate ($\text{Zn}(\text{Ac})_2 \cdot 2\text{H}_2\text{O}$), potassium hydroxide (KOH), dichloromethane (DCM), ethanol, and methanol were supplied by Beijing Tong Guang Fine Chemical Co.,

Ltd. Phenyl triethoxy silane (PhTES) and stearic acid (SA) were provided by Shanghai Aladdin Biochemical Technology Co., Ltd. Polycarbonate (PC) was obtained from Teijin Limited. All of the reagents were of analytical purity and could be used directly.

2.2 Synthesis of monodisperse ZnO nanoparticles

$\text{Zn}(\text{Ac})_2 \cdot 2\text{H}_2\text{O}$ methanol solution (50 mL, 0.33 mol/L) should be added into a flask-3-neck before the reaction starts. Then the KOH ethanol solution (50 mL, 0.66 mol/L) was slowly dripped in the reaction system at 70 °C and then the compound modifier solution (PhTES:SA mass ratio of 14:1) was added dropwise with stirring. After stirring at a constant temperature for 3 h, 10 mL deionized water was added to stop the reaction. After centrifugation, the supernatant was poured away, and then deionized water was added to wash the precipitation. The process was repeated three times. ZnO nanoparticles were obtained after washing and rotary evaporation, and then DCM was added to form a transparent nanodispersion with monodispersed ZnO. In the paper, the modified and unmodified ZnO was represented by MZnO and UZnO, respectively.

2.3 Preparation of ZnO/PC nanocomposites

PC was dissolved in DCM to prepare the PC solution with a concentration of 10 wt %. ZnO nanodispersions with 10 wt % solid content were added slowly to PC solution with constant stirring for 0.5 h to form a high transparent “liquid” masterbatch. Then the “solid” masterbatch was prepared by spray drying. The solid masterbatch containing 5 wt % ZnO was mixed with pure PC for granulation, blended and extruded with a micro-mixing extruder (270 °C, 60 rpm). Meanwhile, nanocomposites containing 1 wt %, 2 wt %, and 3 wt % ZnO were prepared by adding solid masterbatches with 20 wt %, 40 wt %, and 60 wt %, respectively. Test specimens were obtained with a micro-injection molding instrument. Injection molding parameters: injection temperature was 260 °C, mold temperature was 100 °C, and injection pressure was 90 MPa. The fabrication process of ZnO/PC nanocomposites required to prepare “liquid” masterbatch first and “solid” masterbatch later, so the method was called the “liquid–solid masterbatch” blending method.

2.4 Characterization

The micro-morphology and structure of ZnO nanoparticles were obtained with a Hitachi HT7700 Transmission Electron Microscope (TEM) instrument at an accelerating voltage of 100 kV. The crystal structure of ZnO nanoparticles was characterized by an Ultima IV X-ray diffractometer (XRD). The scanning range was from 5 to 80° with a 5°/

min increment. The functional groups on the surface of ZnO were analyzed with Fourier Transform infrared spectroscopy (FT-IR, Vector 70, Bruker). The thermal weight loss of samples was analyzed with thermogravimetric analysis (TGA, STA449C, Netsch) at a heating rate of 10 °C/min in air. Ultraviolet–Visible (UV–Vis) transmittance spectra of ZnO nanoparticles and ZnO/PC nanocomposites were analyzed with UV-2600 UV–Vis spectrophotometer (Shimadzu) in the range 300–800 nm in 0.5 nm increments. EMUC7 ultra-thin microtome was used to prepare TEM section samples of nanocomposites, and the dispersion of ZnO nanoparticles in the PC matrix was analyzed with TEM. The tensile strength and elongation at break of the samples were tested with an electronic universal testing machine (5966, Instron). The single-notch impact strength of the samples was tested with a simple beam impact testing machine (XJJ-5, Hebei Kecheng). The microscopic morphology of the tensile and impact fracture of the samples were analyzed with Scanning Electron Microscope (SEM, S-4800, Hitachi). The molecular weight of ZnO/PC nanocomposite was analyzed with Gel Permeation Chromatography (GPC, PL-GPC50). The photoaging accelerated test was finished in a radiation resistance testing machine bought from Wuxi Silian Science Technology Co., Ltd. The radiation waveband was 290–800 nm. The

peak was at 365 nm, and the maximum radiation intensity can be stabilized at 1200 W/m².

3 Results and discussion

3.1 Modification principle

ZnO nanoparticles have a small size effect, which makes them easy to agglomerate in large quantities. To improve the dispersion and the compatibility of the ZnO nanoparticles in the PC matrix, ZnO nanoparticles were modified and monodispersed in DCM, a suitable solvent for PC resin. Two modifiers with different structures were mixed and grafted onto the surface of ZnO nanoparticles. One was phenyl triethoxy silane, and the other was stearic acid.

The process of grafting the modifiers on the surface of ZnO nanoparticles is presented in Fig. 1. The simple structural formula of phenyl triethoxy silane is C₁₂H₂₀O₃Si, which can be outlined in the formula of RSiX₃. The alkoxy groups of C₁₂H₂₀O₃Si hydrolyze form new functional silanol groups, which may bind the hydroxyl groups of ZnO nanoparticle surface by hydrogen bonds [31]. Meanwhile, the stearic acid molecule is a chain-like structure with one end

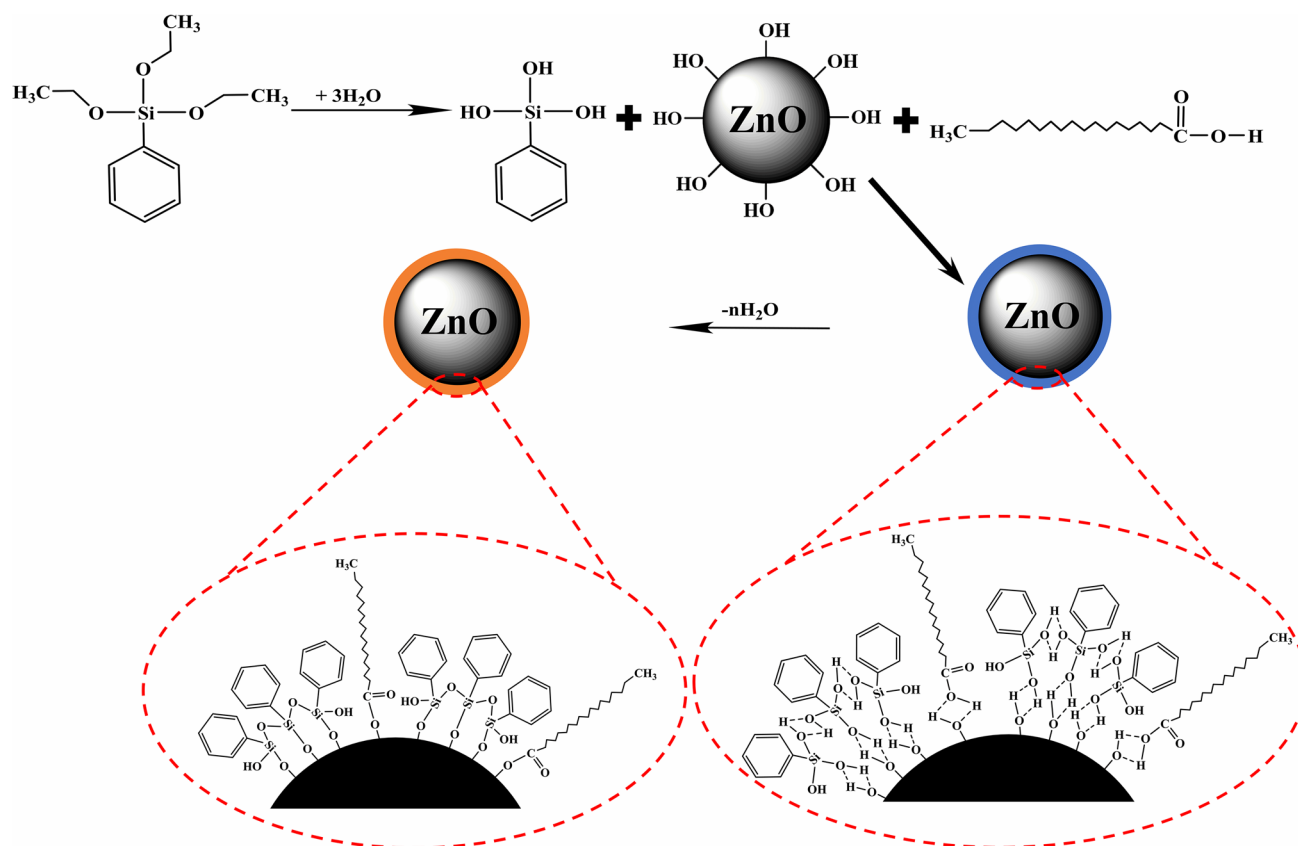


Fig. 1 Schematic diagram of grafting modifier onto ZnO nanoparticle

of a hydrophilic group, which can react with the hydroxyl group on the surface of ZnO nanoparticles to form a chemical bond. The just forming reaction products are dehydrated in some conditions, and the final products are obtained [32]. Thus, the phenyl triethoxy silane and stearic acid are successfully grafted on the surface of ZnO nanoparticles. There are a lot of long alkyl chain structures and phenyl groups on the surface of ZnO, which cannot only greatly enhance the steric effect between ZnO nanoparticles but also improve the compatibility of ZnO in the dispersed media, contributing to the achievement of a fine dispersion of ZnO nanoparticles in DCM and PC matrix.

3.2 Morphology and structure of monodispersed ZnO nanoparticles

Figure 2a, b shows the TEM image of ZnO nanoparticles and the UV–Vis transmittance of the nanodispersions with 10 wt % ZnO nanoparticles in DCM, respectively. ZnO has an average particle size of about 6 nm and is monodispersed in DCM. The organic chains of modifiers are attached to the surface of ZnO nanoparticles, which can enhance the dispersion of nanoparticles in the target solvent DCM, as mentioned above. The excellent dispersion state leads to the excellent optical properties of nanodispersions. From the UV–Vis transmission spectrum curve, it can be seen that the nanodispersions of ZnO in DCM can completely block the ultraviolet light below 380 nm, while its visible light transmittance reaches 92% at 550 nm, suggesting no Rayleigh scattering caused by the aggregations of nanoparticles. Therefore, the nanodispersions may hold high transparency. The digital photograph of ZnO nanodispersions is shown in the inset picture of Fig. 2b. It is colorless and transparent, further proving the excellent dispersion of ZnO nanoparticles in DCM. Moreover, the nanodispersions exhibit excellent dispersion stability, with no precipitation after the storage for 6 months.

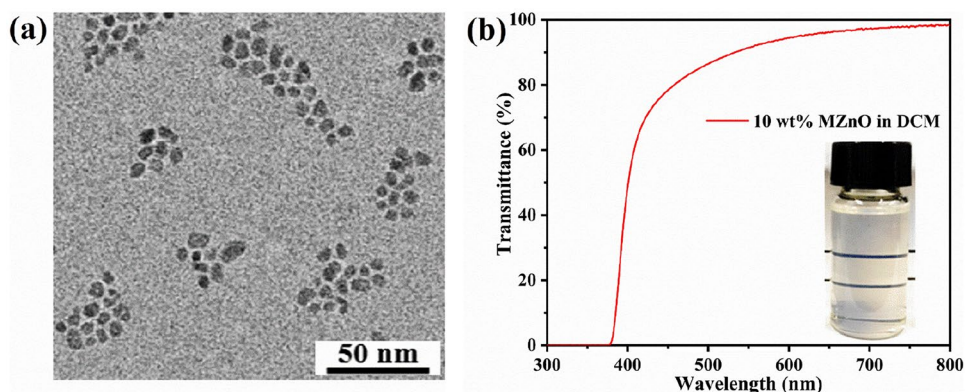
The XRD spectra of UZnO nanoparticles and MZnO nanoparticles are shown in Fig. 3a. MZnO and UZnO samples

have diffraction peaks at 2θ of 31.7° , 34.4° , 36.2° , 47.5° , 56.6° , 62.8° , and 68.0° . By comparing the data with JCPDS No.361451, it shows that these diffraction peaks belong to the (100), (002), (101), (102), (110), (103), and (112) crystal surfaces of ZnO with hexagonal wurtzite structures, respectively. In addition, compared with UZnO, the diffraction peak area of MZnO is broadened, and the intensity is weakened, which further confirms that many organic modifiers are grafted on the surface of MZnO [33].

The chemical structure evolution of SA, PhTES, MZnO, and UZnO nanoparticles is investigated by FTIR analysis (Fig. 3b). The peaks at 480 cm^{-1} and 3400 cm^{-1} on the infrared spectrum curves of MZnO and UZnO belong to the stretching vibration peaks of Zn–O and O–H bonds [20]. The peaks of PhTES and MZnO at 1110 cm^{-1} are the vibration peaks of Si–O–Si, and the vibration peaks of the H on phenyl are at 694 cm^{-1} and 735 cm^{-1} [34]. On the curve of SA, the vibration peak of the carboxylic acid carbonyl group (C=O) is at 1705 cm^{-1} , while it disappears and the new double vibration peaks at 1430 cm^{-1} and 1550 cm^{-1} appear on the curve of MZnO. Because SA reacts with the hydroxyl groups on the surface of ZnO, the C=O vibration peak of SA transforms to the symmetrical and antisymmetrical vibration peaks of the bidentate conjugate mode [35]. In addition, C–H bond vibration peaks of $-\text{CH}_2$ and $-\text{CH}_3$ that belong to SA appear at 2850 cm^{-1} and 2918 cm^{-1} . As mentioned above, it can be concluded that the modifiers (PhTES and SA) have been successfully grafted on the surface of ZnO nanoparticles.

The TG and DTG curves of UZnO and MZnO nanoparticles are illustrated in Fig. 3c. The total weight loss of UZnO is 3 wt %, which is mainly due to the decomposition of adsorbed water and hydroxyl groups on the surface of the nanoparticles. The maximum weight loss rate is reached at the temperature of 200°C , and the primary removal is the adsorbed water on the surface of the nanoparticles. Above this temperature, the hydroxyl group begins to be removed from the surface until it is completely decomposed [36]. For MZnO, the weight loss rate before 290°C is equivalent to that of UZnO. The mass loss of adsorbed water and a

Fig. 2 **a** TEM image of MZnO; **b** UV–Vis transmission spectrum of 10 wt % MZnO nanoparticles in DCM (inset picture shows the digital photograph of ZnO nanodispersions)



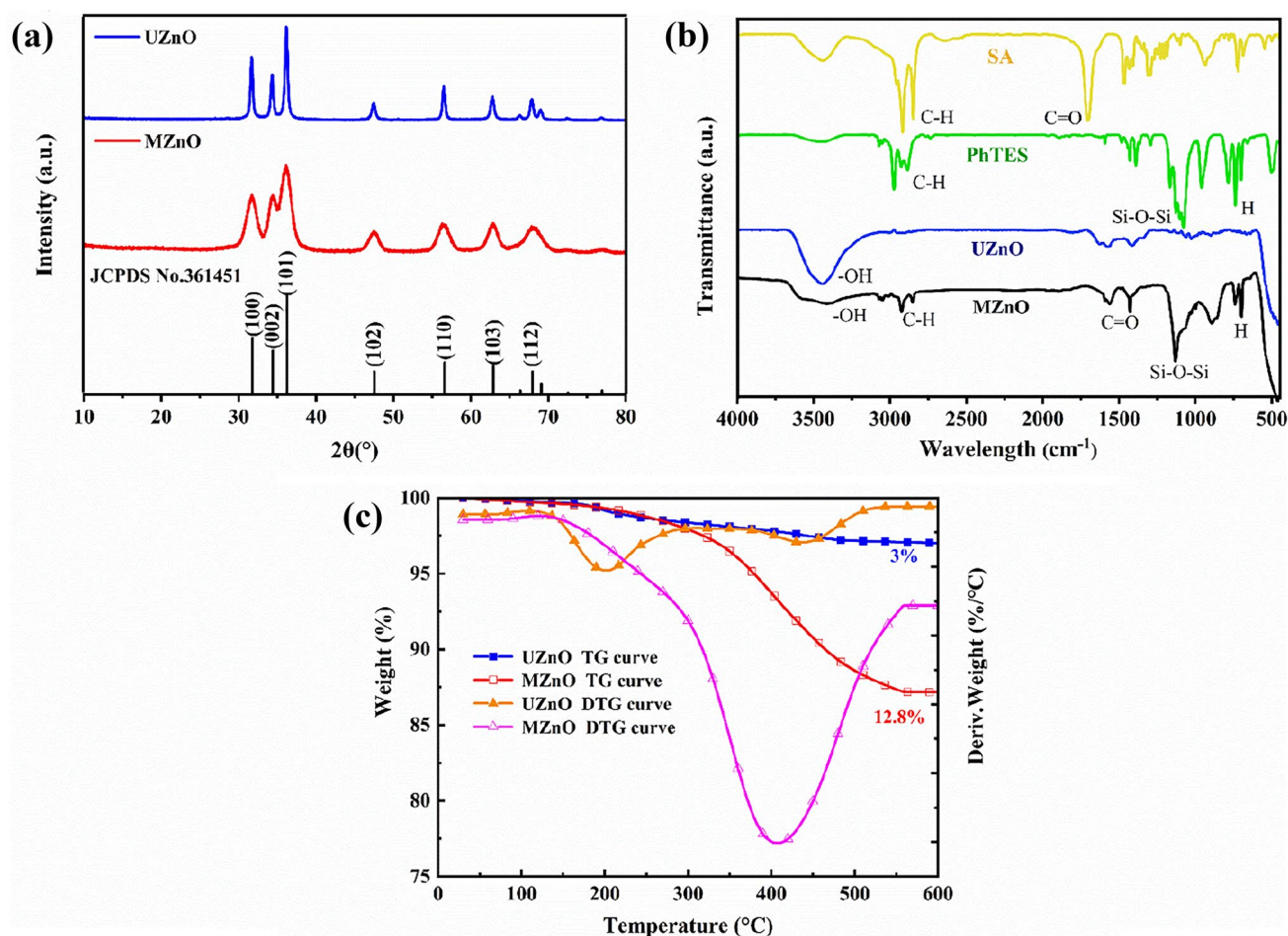


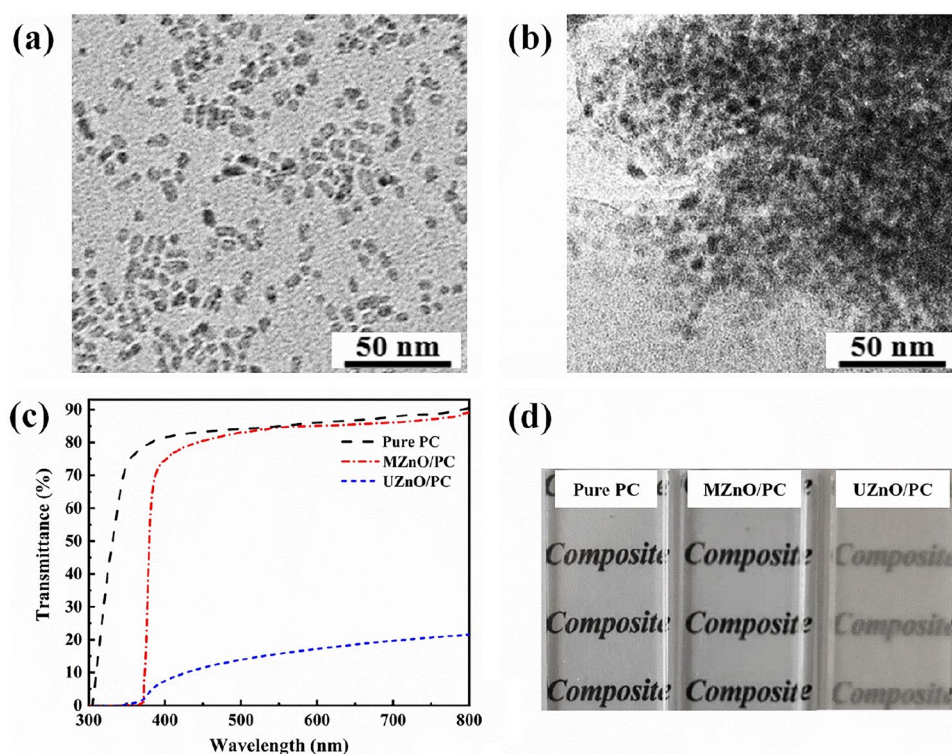
Fig. 3 **a** XRD spectra of UZnO and MZnO; **b** FTIR spectra of SA, PhTES, UZnO, and MZnO; **c** TG and DTG curves of UZnO and MZnO

small number of hydroxyl groups is about 1.5 wt %. After 290 $^{\circ}\text{C}$, with the increase of temperature, the weight loss gradually increases. MZnO has the maximum weight loss rate at 406 $^{\circ}\text{C}$ and is decomposed completely at 558 $^{\circ}\text{C}$. The grafted organic modifiers on the surface of particles account for about 9.8 wt % of the total weight. The TG and DTG results show that the modifiers on MZnO have good thermal stability and are not decomposed at 250–300 $^{\circ}\text{C}$, which is the melt processing temperature of PC [37]. The modifiers on the surface of ZnO nanoparticles have phenyl groups and saturated long-chain alkyl groups, which have good thermal stability. Moreover, the modifiers react with hydroxyl groups on the surface of nanoparticles to form stable chemical bonds. Therefore, the surfactants of nanoparticles are not easy to be decomposed during the melt processing, and the prepared nanocomposites do not become yellow. In the reported literatures [22, 33, 38–40], most modified nanoparticles generally begin to be decomposed below 250 $^{\circ}\text{C}$, which cannot be incorporated into the thermal plastics with high processing temperature, such as PC, polysulphone, and polyethylene terephthalate.

3.3 Optical properties of ZnO/PC nanocomposites

MZnO nanoparticles are uniformly dispersed in the PC matrix and present a monodisperse state, as demonstrated by the TEM image of nanocomposites in Fig. 4a. In contrast, UZnO nanoparticles exist as the large agglomerates in the PC matrix (Fig. 4b). In general, ZnO nanoparticles are hydrophilic, making them difficult to be uniformly dispersed in organic polymers. The modifiers play the role of “bridges” between inorganic fillers and organic polymers to enhance the binding between ZnO nanoparticles and PC, significantly improving ZnO dispersion in the PC matrix. Such a homogenous dispersion makes the nanocomposites achieve excellent comprehensive properties. Especially for the optical PC plastic, the addition of ZnO nanoparticles does not affect its visible light transparency, as revealed by the UV–Vis transmission spectra curves in Fig. 4c. MZnO/PC nanocomposite maintains the same visible light transmittance as pure PC, and the UV shielding ability of PC can be significantly improved by ZnO. When the content of ZnO is 1 wt %, MZnO/PC nanocomposite has a visible light

Fig. 4 TEM images of **a** MZnO/PC and **b** UZnO/PC nanocomposites; **c** UV–Vis transmission spectra of pure PC, MZnO/PC, and UZnO/PC nanocomposites; **d** digital photos of pure PC, MZnO/PC, and UZnO/PC nanocomposites (the content of ZnO is 1 wt %)



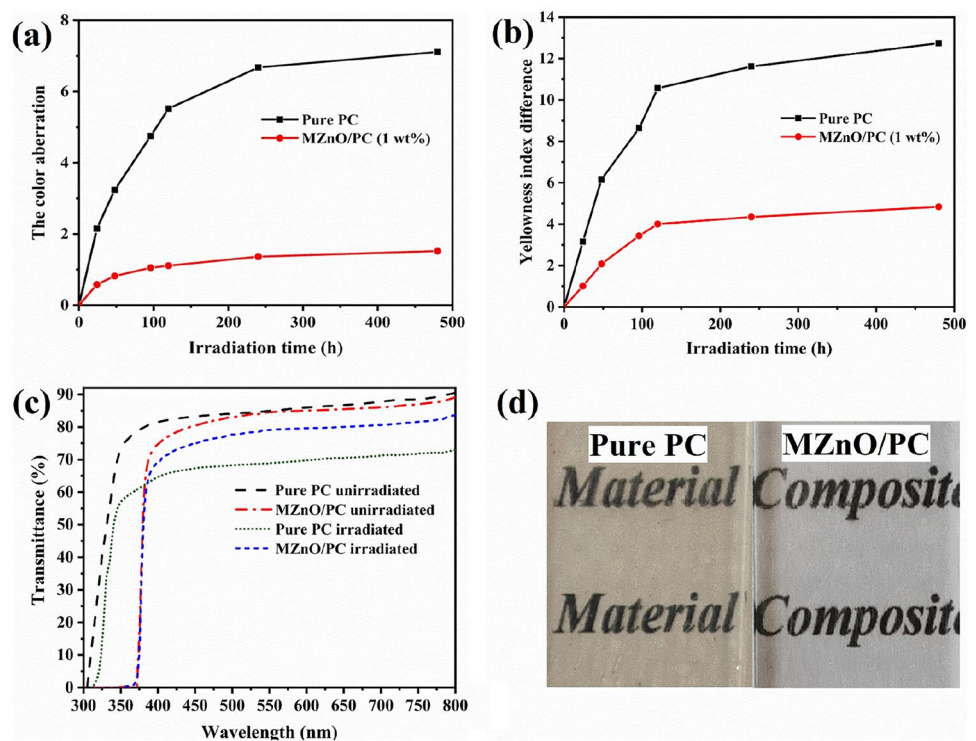
transmittance of 84% at 550 nm and can completely block ultraviolet below 370 nm. It further confirms that MZnO has good dispersibility in the PC matrix, and the nanocomposites can keep transparency. In addition, the digital photos of pure PC, MZnO/PC, and UZnO/PC nanocomposites (Fig. 4d) also exhibit that MZnO/PC nanocomposite is colorless and transparent, which are consistent with the appearance of pure PC. It has not been yellowed, owing to the good thermal stability of the surface modifiers as mentioned above. But with the addition of UZnO nanoparticles, the visible light transmittance at 550 nm of the nanocomposite is significantly reduced from 85 to 18%. UZnO/PC nanocomposite has become translucent, indicating that UZnO nanoparticles are seriously agglomerated in the PC matrix, causing the Rayleigh scattering and a noticeable decrease in the visible light transmittance.

PC as an organic polymer material can be degraded and destroyed under the action of ultraviolet. At this point, its appearance changes to yellow, and the transmittance of visible light is decreased. ZnO nanoparticles with excellent UV shielding property protect the objects and people from UV damage and improve the light aging resistance of transparent polymer matrices. Different test samples of pure PC and the nanocomposites with 1 wt % MZnO are irradiated by ultraviolet to investigate the changes in the optical performance. The effects of MZnO on the color aberration, yellowness index, UV–Vis's transmittance, and

appearance of the composites are shown in Fig. 5. With the irradiation of ultraviolet, the color aberration and yellowness index of pure PC change greatly (the black curves in Fig. 5a, b). After 480 h irradiation on pure PC, the color aberration and the yellowness index reach 7.1 and 12.8, respectively. Pure PC undergoes the “Photo-Fries” reaction when irradiated by ultraviolet [41]. It generates molecules with biphenyl group structure, which can absorb visible light, resulting in the change in the color of the materials [42]. Under the protection of MZnO, the nanocomposites have a slight color change. The color aberration and yellowness index are 1.5 and 4.8 after accelerated photo-aging for 480 h, which are less than pure PC.

The photo-aging resistance properties of samples are also evaluated using a UV–Vis transmittance test, and the results are shown in Fig. 5c. After 480 h of UV irradiation, the transmittance of the nanocomposites at 550 nm in the visible light region only is slightly decreased by 5% (from 84 to 79%). However, the transmittance of pure PC is significantly reduced by 16% (from 84 to 68%). From the digital photos of pure PC and MZnO/PC nanocomposite after UV irradiation (Fig. 5d), it can also be seen that pure PC yellows obviously. In contrast, MZnO/PC nanocomposite remains the original state without any change, which further confirms that ZnO can prevent PC from undergoing the “Photo-Fries” reaction and improve the photo-aging resistance of PC remarkably.

Fig. 5 **a** Relation of color aberration and irradiation time of samples; **b** the relationship between different irradiation time of samples and the difference of yellowness index; **c** UV–Vis transmission spectra of the samples before and after aging; **d** digital photos of pure PC and MZnO/PC nanocomposite after UV irradiation



3.4 Mechanical properties of ZnO/PC nanocomposites

The influence of ZnO content on the mechanical properties of nanocomposites is illustrated in Fig. 6a. With the increase of ZnO content, the tensile strength, elongation at break, and impact strength of nanocomposites all show the trend of first increasing and then decreasing. When the additive amount of ZnO is 1 wt %, the nanocomposite has the best comprehensive mechanical properties. The tensile strength, elongation at break, and single-notch impact strength reached 66 MPa, 72%, and 53 kJ/m², respectively. However, with the addition of MZnO to excess, the relative density of nanoparticles in the matrix will increase. When the nanocomposite is subjected to external forces, a large number of micro-cracks and excessive plastic deformation will appear, and the ductility flow of the polymer chain will be limited, so that the impact strength, tensile strength, and elongation at break of the composite begin to decline and no longer maintain the excellent mechanical properties [43–45]. The mechanical performances of pure PC, UZnO/PC, and MZnO/PC nanocomposites with 1 wt % ZnO are summarized in Fig. 6b. It can be found that the tensile and impact properties of MZnO/PC nanocomposite are significantly superior to those of pure PC and UZnO/PC nanocomposite. Compared with pure PC, the tensile strength, elongation at break, and impact strength of MZnO/PC nanocomposite have enhanced by 10%, 25%, and 6%. However, the mechanical properties of UZnO/PC

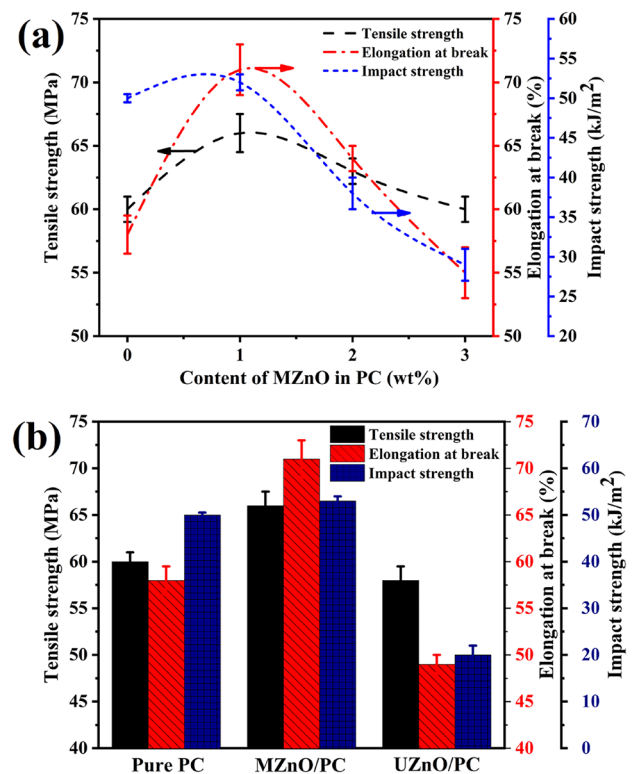


Fig. 6 **a** Mechanical properties of MZnO/PC nanocomposites with different solid contents; **b** mechanical properties of pure PC and nanocomposites (the content of ZnO is 1 wt %)

composite are lower than that of pure PC because of the aggregation of ZnO nanoparticles in the PC matrix.

Figure 7a–c shows the SEM images of tensile fracture surfaces of pure PC, MZnO/PC, and UZnO/PC nanocomposites. The tensile fracture surface of pure PC is flat, while those of ZnO/PC nanocomposites are rough. Compared with MZnO/PC nanocomposite, the number of microvoids on the fracture surface of UZnO/PC nanocomposite is fewer, and the roughness of the fracture surface is smaller. There are relatively deep and big grooves on the fracture surface of UZnO/PC nanocomposite, as shown in the white arrow in Fig. 7c. During the stretching process, the stress transfers from the matrix to the filler, and its transfer efficiency is affected by the interface area and interface adhesion. MZnO nanoparticles are well-dispersed in PC and have a solid binding force with PC due to modifiers, which enlarges the interface contact area and enhances the interface adhesion, which is of great help to the stress dissipation. When the interface between the nanoparticles and the matrix is separated under external force, many homogeneously dispersed microvoids are formed, and the tensile fracture surface becomes rough, which helps to absorb stress evenly and improve the tensile strength [46]. However, as for UZnO nanoparticles, the interface interaction between them and the polymer matrix is weak, resulting in that they are easily falling off from

the matrix, which hinders effective stress transmission and dissipation. Moreover, because UZnO nanoparticles aggregate seriously in the matrix, deeper and larger grooves can be generated on the interface during the stretching process, which will initiate the large cracks and decrease the tensile strength of nanocomposites.

As evident from SEM images of impact fracture surfaces of pure PC, MZnO/PC, and UZnO/PC nanocomposites (Fig. 7d–f), pure PC sample forms a few yield shear bands under the action of the external impact force and discontinuous periodic fringes between the shear bands perpendicular to the direction of fracture propagation, as given in the white box in Fig. 7d, which shows a typical ductile fracture surface. Compared with pure PC, the impact fracture surface of MZnO/PC nanocomposite has denser and longer yield shear bands. Continuous and more white periodic stripes between the shear bands are formed, as shown in the box in Fig. 7e. Uniformly dispersed nanoparticles acting as the points of stress concentration cause the surrounding polymer matrix to produce denser and longer yield shear bands and trigger a large number of crazes between adjacent shear bands during the impact process, which consumes a large amount of energy and significantly improves the impact strength of the materials [47]. Compared with pure PC, the fracture surface of the nanocomposite has more shear bands and crazes, so

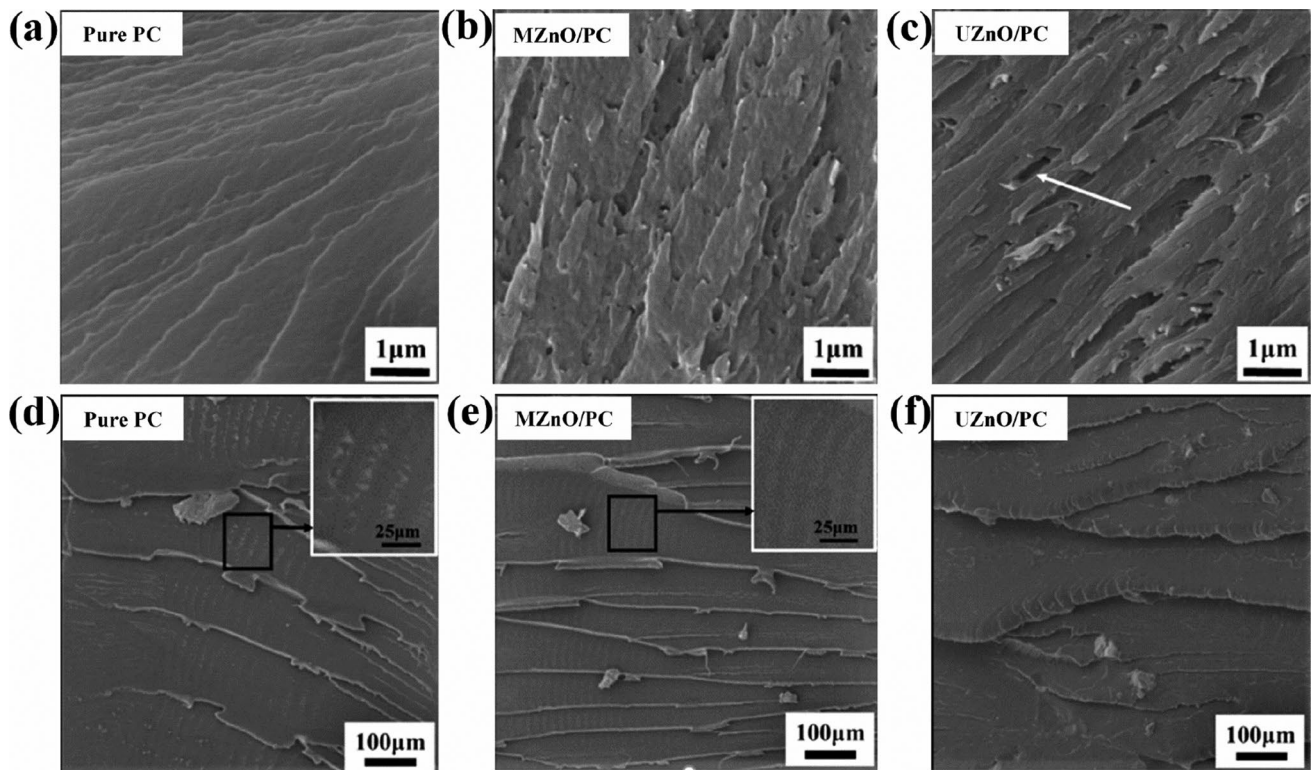


Fig. 7 SEM images of tensile fracture surfaces of **a** pure PC, **b** MZnO/PC, and **c** UZnO/PC and impact fracture surfaces of **d** pure PC, **e** MZnO/PC, and **f** UZnO/PC

the nanocomposite has better impact toughness. However, UZnO/PC nanocomposite does not show yield shear bands and periodic fringes, and the cross-section shows a typical brittle fracture with many cracks. The large aggregates of nanoparticles in the matrix cause the materials to undergo a tough-brittle transition, resulting in a significant decrease in the impact strength [44].

The photo-aging of PC affects its optical performance and has a considerable effect on its mechanical properties. It becomes hard and brittle because the molecular weight is reduced [48], and many micro-cracks are produced under the radiation of ultraviolet lights [49, 50]. The relationships between the irradiation time and the mechanical properties of pure PC and MZnO/PC nanocomposite are illustrated in Fig. 8a–c. With the extension of irradiation time, the tensile strength, elongation at break, and impact strength of the samples are declined to a different degree. Compared with pure PC, the MZnO/PC nanocomposite has a slighter decrease in strength and toughness. After 480 h

UV irradiation, the tensile strength and elongation at break of pure PC are reduced by 8.3% and 6.8%, while that of MZnO/PC is decreased by only 2.5% and 2.8% (Fig. 8a, b). More importantly, the toughness of nanocomposites irradiated by UV rays is improved by ZnO, as shown in Fig. 8c. After 480 h UV irradiation, the impact strength of MZnO/PC nanocomposite is declined to 45 kJ/m² from 52 kJ/m², decreasing by 13.5%. The decline extent of impact strength is smaller than that of pure PC, which is reduced by 44.0% (from 50 to 28 kJ/m²).

The mechanical properties of polymers are directly related to their molecular weight. The molecular weight and distribution of pure PC and MZnO/PC nanocomposite before and after UV irradiation are given in Table 1. The weight-average molecular weight and number-average molecular weight are represented by M_w and M_n , respectively, and the molecular weight distribution (D) is the value of M_w/M_n . The molecular weights of both pure PC and MZnO/PC nanocomposite decreased after UV irradiation for 480 h. M_w and

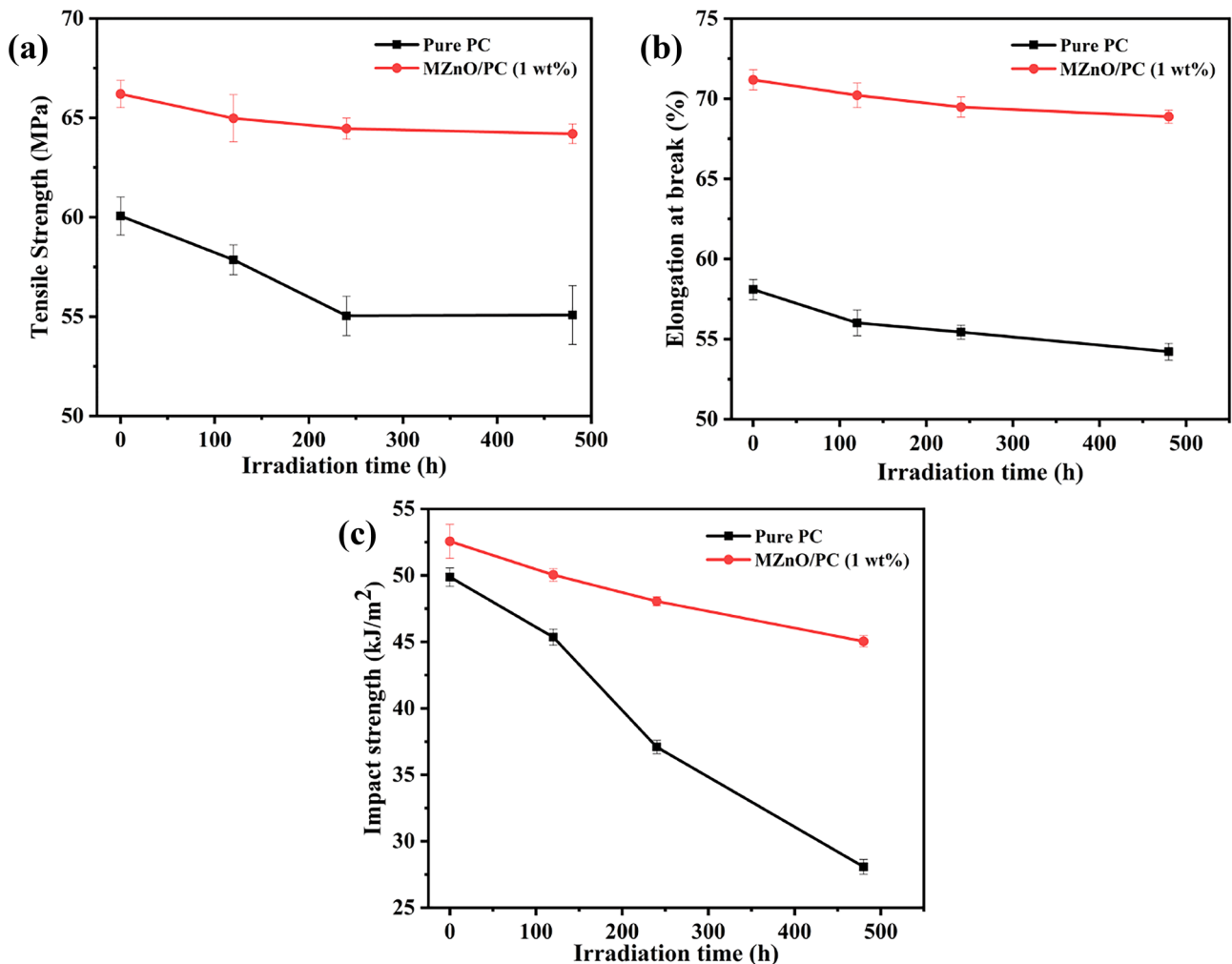


Fig. 8 a Tensile strength, b elongation at break, and c impact strength of pure PC and MZnO/PC nanocomposite after UV radiation

Table 1 The molecular weight and distribution of samples before and after UV irradiation

Time (h)	Pure PC			MZnO/PC (1 wt %)		
	$M_w \times 10^4$	$M_n \times 10^4$	D (M_w/M_n)	$M_w \times 10^4$	$M_n \times 10^4$	D (M_w/M_n)
0	5.32	2.46	2.16	5.24	2.41	2.17
480	4.96	2.21	2.24	5.13	2.38	2.16

M_n of pure PC are 6.8%, 10.2% lower than before irradiation, and the value of D is significantly widened from 2.16 to 2.24. However, M_w and M_n of MZnO/PC are only 2.1%, 1.2% lower than before irradiation in the same aging time, and D is almost unchanged. Therefore, from the changes of molecular weights before and after UV irradiation, it can be directly proved that ZnO can slow down the fracture of the molecular chain and decrease the mechanical properties under UV irradiation.

4 Conclusions

A highly transparent ZnO/PC nanocomposite with excellent comprehensive performances was successfully fabricated by the “liquid–solid masterbatches” blending method. The synthesis of monodispersed ZnO nanoparticles and preparation of “liquid” and “solid” masterbatches contribute to the achievement of uniform distribution of ZnO nanoparticles in the PC matrix, resulting in the excellent optical properties, mechanical performances, and photo-aging resistance of ZnO/PC nanocomposite. ZnO nanoparticles with an average particles size of about 6 nm were modified by two surfactants with good thermal stability and monodispersed in DCM, which are suitable for PC plastic with high processing temperature. ZnO/PC nanocomposite had the same visible light transmittance as pure PC while completely blocking harmful ultraviolet below 370 nm. The tensile strength, elongation at break, and impact strength of the nanocomposite were increased by 10%, 25%, and 6% with 1 wt% ZnO. The strength and toughness of the materials were simultaneously improved. More importantly, ZnO can improve the UV anti-aging performance of composites remarkably. After 480 h of UV irradiation, compared with pure PC, the visible light transmittance at 550 nm of ZnO/PC nanocomposite was decreased by only 5% from 16%. The impact strength of ZnO/PC nanocomposite was declined by 13% from 44%. The weight average molecular weight and number average molecular weight were slightly reduced, and the color of the samples was kept unchanged. This study has significant guiding significance and reference value for preparing thermal-plastic-based optical inorganic/organic nanocomposites, which have a great potential application in such fields as aeronautical and space, electronic information, and medical treatment.

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Declarations

Conflict of interest The authors declare no competing interests.

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